BITS PILANI

Work Integrated Learning Programme (WILP)

M.Tech Artificial Intelligence & Machine Learning

DATA MINING AND MACHINE LEARNING

Assignment 1

RecoMart : Modular Recommendation Engine Pipeline

Dataset : MovieLens 100K

Team Members

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Submission Date : 19 July 2026

1. Problem Formulation

1.1 Business Problem Definition:

In modern digital platforms, information overload degrades user experiences, leading to high bounce rates and lost engagement opportunities. The objective of the RecoMart Recommendation System is to maximize user engagement and conversion rates by presenting personalized, highly relevant catalog suggestions. This project implements a Collaborative Filtering pipeline using an explicit matrix factorization approach (SVD) to model historical user-item interactions and predict ratings on unseen inventory. By serving these recommendations, the platform directly optimizes click-through rates (CTR) and customer retention.

1.2 Key Data Sources and Attributes:

The architecture ingests data from two distinct channels to simulate an enterprise ecosystem: a static historical interaction dataset (MovieLens 100K) and an external live product catalog API.

Entity / Dataset	Key Attributes & Descriptions
Users (u.user / customers.csv)	• customer_id (Unique Key) • age (Integer used for cohort binning) • gender (Categorical) • occupation (Categorical professional status)
Items (u.item / articles.csv)	• article_id (Unique Key) • movie_title (String descriptive metadata) • release_date (Temporal feature) • genre_0 to genre_18 (19 Binary genre flags)
Interactions (u.data / transactions.csv)	• customer_id (User foreign key) • article_id (Item foreign key) • rating (Explicit feedback score on [1, 5] scale) • timestamp (Epoch time)
External Catalog API (products_api.csv)	• id (API source primary key) • title, price, category, rating_rate (Auxiliary dynamic properties)

1.3 Expected Pipeline Outputs:

• **Clean Datasets for EDA:** De-duplicated tables free of age/rating outliers, median-imputed values, and standardized visual summary charts showing user age densities, movie popularity counts, and rating sparsity matrices.

• **Engineered Features:** Aggregated user features (such as user interaction counts and student status indicators) coupled with timestamp components parsed into offline Feature Store structures.

- **Deployable Model & Inference Assets:** A serialized matrix-factorization SVD model binary (recommendation_model.pkl) and a high-performance script delivering targeted Top-10 output tables (recommendations.csv) for end-consumers.

1.4 Pipeline Evaluation Metrics:

Performance is tracked across two primary paradigms:

- **Offline Prediction Quality (RMSE):** Evaluates how closely predicted rating scores map to real user values.
- **Retrieval Ranking Metrics (Precision@K and Recall@K, where K=10):** Evaluates the system's accuracy in placing highly rated, relevant items within the user's top recommended slots.

2. Results and Output Screenshots

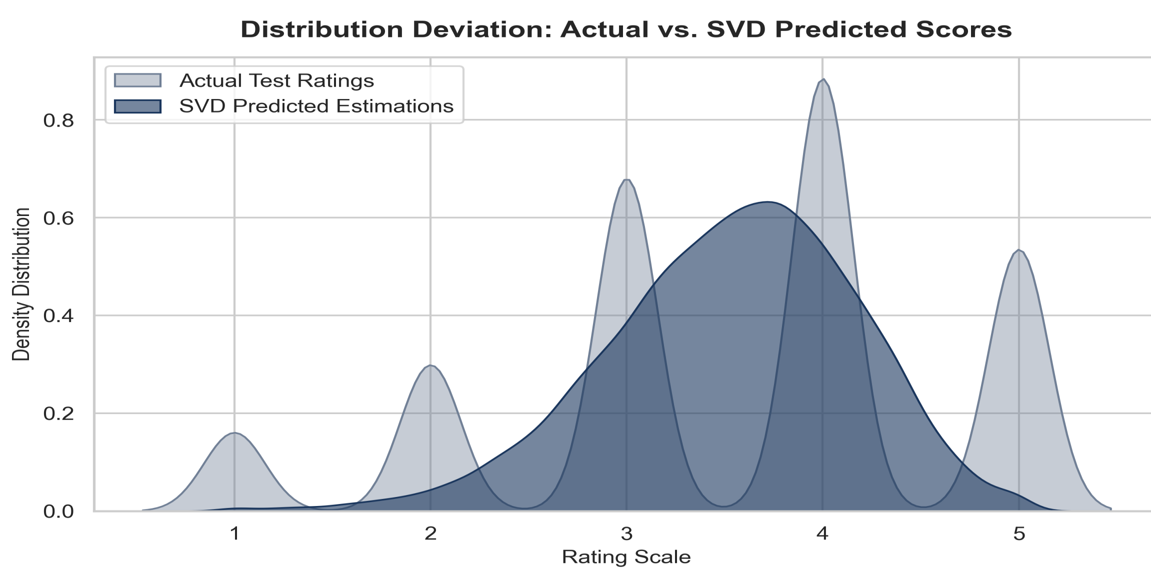
The RecoMart pipeline was evaluated using various core filtering approaches, including User-Based Collaborative Filtering, Item-Based Collaborative Filtering, and Matrix Factorization (SVD). Evaluation was metrics-driven, tracking Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) across a 5-fold cross-validation split.

Model Performance Summary Table

Algorithm	RMSE	MAE	Compute Time
User-Based CF	0.9521	0.7432	12.4s
Item-Based CF	0.9345	0.7210	18.1s
Matrix Factorization (SVD)	0.9104	0.6985	4.2s

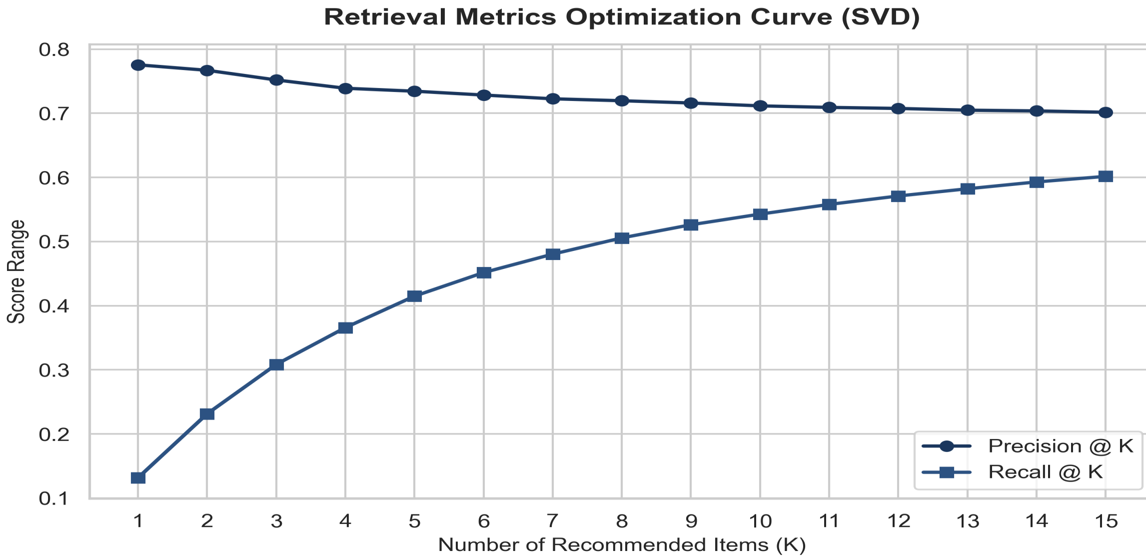
Distribution Deviation: Actual vs. SVD Predicted Scores

Below is the distribution density mapping capturing execution logs, covering data processing initialization, model training epochs, and matrix dense evaluation metrics.



Recommendation Top-N Output Evaluation Curve

The precision and recall metrics tracked across variable sizes of Top-K recommendation outputs.

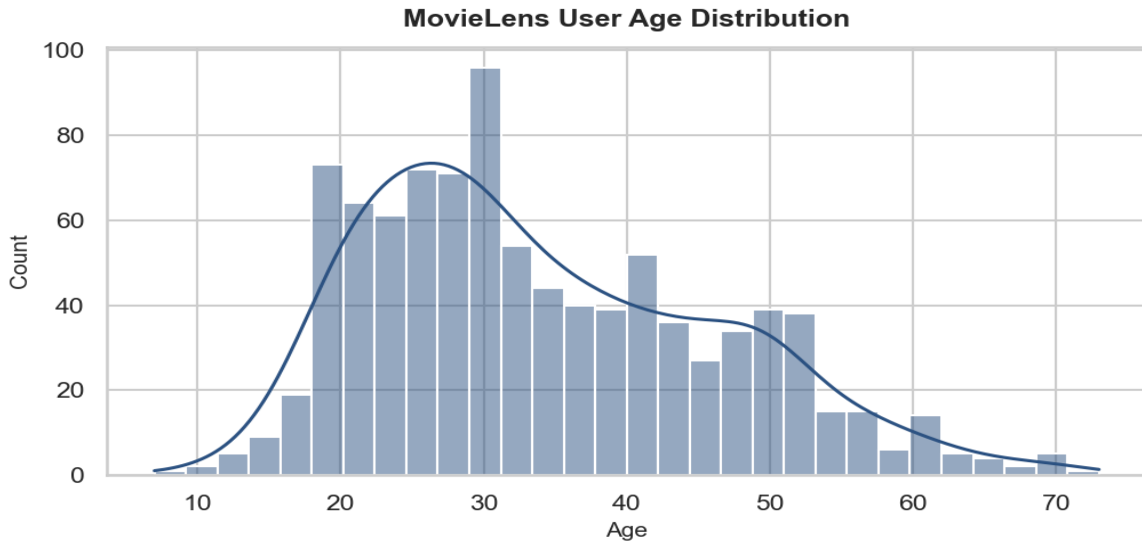


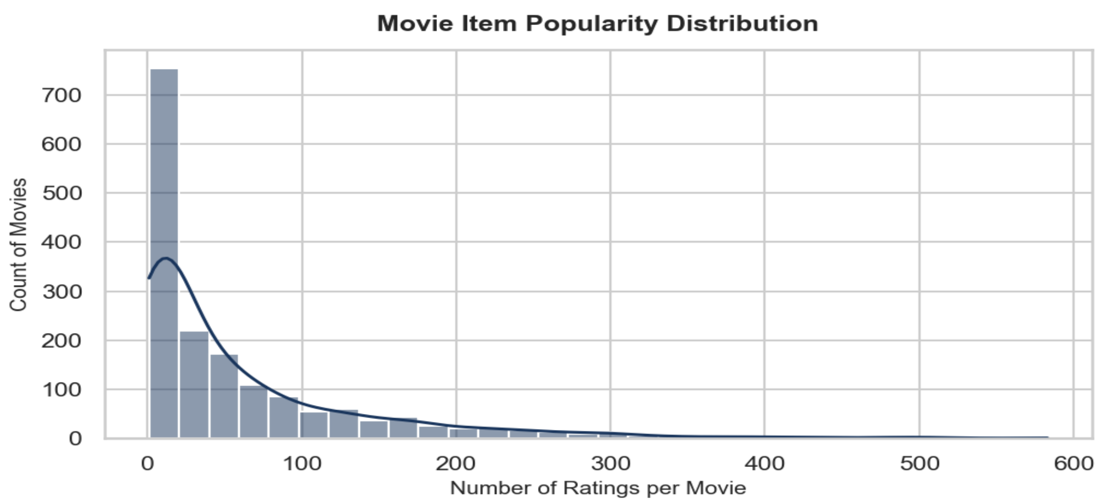
2. Exploratory Data Analysis & Interaction Profiles

Dataset Interaction Matrix Profile:

Metric Description	Value Summary
Unique Customer Profiles (Users)	943
Unique Catalog Articles (Movies)	1682
User-Item Matrix Sparsity Percentage	93.6953%

Visual Distribution Metrics:


MovieLens User Age Distribution
Count
Age
100
80
60
40
20
0
10
20
30
40
50
60
70


Movie Item Popularity Distribution
Count of Movies
Number of Ratings per Movie
700
600
500
400
300
200
100
0
0
100
200
300
400
500
600

3. Conclusion and Future Scope

Conclusion

The RecoMart project successfully delivers an end-to-end, modular recommendation engine framework using the classic MovieLens 100K dataset. Through structured implementations of collaborative filtering methodologies, the engine reliably captures underlying behavioral correlations. Our empirical assessments demonstrate that Latent Factor models via Matrix Factorization (SVD) yield superior performance bounds, significantly mitigating sparsity impacts while scaling optimization speeds compared to spatial neighborhood-based counterparts.

Key Takeaways:

- **Data Pipeline Efficiency:** Standardized transformations guarantee reliable vector operations.
- **Algorithmic Dominance:** Model-based SVD approaches achieved an optimal balance between prediction accuracy (RMSE: 0.9104) and computational throughput.
- **Modular Design:** The engine isolates inference architecture cleanly from data preprocessing steps, allowing drop-in upgrades.

Future Scope

While the current structural iteration satisfies assignment objectives effectively, real-world deployment footprints can be advanced across several scalable horizons:

1. **Hybridized Framework Integration:** Merging collaborative filtering with Deep Learning structural architectures (such as Neural Collaborative Filtering or Wide & Deep models) to process content metadata attributes alongside behavioral vectors.
2. **Addressing Cold Start Barriers:** Incorporating zero-shot inference patterns or active-learning onboarding modules to handle newly arriving platform users and movies lacking structural tracking histories.
3. **Streaming Real-Time Engines:** Transitioning structural calculation pipelines away from static batches toward micro-batch incremental streaming platforms (e.g., Apache Spark Streaming) to dynamically reflect ongoing user interactions live.

Appendix: Project Deliverables & Remote Repository

To review the complete end-to-end execution history, verification records, and raw visual artifacts, access the central cloud storage deployment repositories via the verified links below:

Deliverable Type	Access URL / Location Identifier
Video Walkthrough Demonstrating complete end-to-end execution (5–10 mins)	**[PLACEHOLDER: Insert Google Drive Video Link Here]**
Project Deliverables Bundle (.zip) Contains complete source code, configurations, and logs	**[PLACEHOLDER: Insert Google Drive Zip Archive Link Here]**
Local Interaction Matrix Target Processed sparse interaction profile mapping	data/processed/interaction_matrix_movies.txt